Assignment 2

Question 1: A company is manufacturing metal rods, and the target length TTT of the rods is 100 mm. The deviation from the target is associated with a loss of ₹50 for a deviation of 5 mm. What is the value of k in the Taguchi loss function? Once the value of k is determined, calculate the loss if a rod is produced with a length of 103 mm.

Question 2: Given the same Taguchi loss function from **Question 1**, calculate and compare the loss if: Rod 1 has a length of 102 mm, Rod 2 has a length of 98 mm, Rod 3 has a length of 101 mm. Which rod has the lowest loss, and by how much?

Question 3: A company can make a simple adjustment at the factory for ₹5 to bring a product's length to 100 mm (the target value). Currently, the product length is 106 mm, and the company incurs a loss of ₹72 for each such deviation. Using the Taguchi Loss Function, calculate the loss for the deviation and determine if making the ₹5 adjustment is economically justified.

Question 4: Two manufacturing processes, A and B, are being considered to produce a component with a target dimension of 200 mm. In Process A, the average deviation is 2 mm, while in Process B, the average deviation is 3 mm. The loss for a 2 mm deviation is ₹25. Assuming the Taguchi Loss Function applies, calculate the loss for both processes and recommend which process the company should use to minimize loss.

Question 9: A process produces parts that deviate by an average of 4 mm from the target value. The Taguchi Loss Function for the process has a cost coefficient of $k = 5$. Calculate the loss per part and determine the total loss if 500 parts are produced.